

Booth Radix-4

DP8 (8×4) — decomposition over the Booth digits of B

Step 1 — the dot product

$$\sum_{i=0}^7 A_i B_i$$

Step 2 — recode B into radix-4 digits and exchange the sums

Each B_i is a 4-bit value read as signed; an overlapping 3-bit window gives one digit per bit pair, $B_i = \sum_{j=0}^1 4^j d_i^j$ with $d_i^j = -2 B_i^{2j+1} + B_i^{2j} + B_i^{2j-1} \in \{-2, \dots, 2\}$, so

$$\sum_{i=0}^7 A_i B_i = \sum_{i=0}^7 \left[\sum_{j=0}^1 4^j d_i^j \right] A_i = \sum_{j=0}^1 4^j \sum_{i=0}^7 d_i^j A_i$$

The bit below the nibble, B_i^{-1} , is 0 for a whole 4-bit operand and the top bit of the nibble below when B_i is a slice of a wider B (Step 5).

Step 3 — the partial product

$$\sum_{i=0}^7 A_i B_i = \sum_{j=0}^1 4^j \sum_{i=0}^7 \underbrace{\left[d_i^j A_i \right]}_{R(B_i^{2j+1}, B_i^{2j}, B_i^{2j-1})}$$

$R(B_i^{2j+1}, B_i^{2j}, B_i^{2j-1})$ depends on the three bits of the window and takes one of five values:

B_i^{2j+1}	B_i^{2j}	B_i^{2j-1}	d_i^j	R
0	0	0	0	0
0	0	1	+1	A_i
0	1	0	+1	A_i
0	1	1	+2	$2A_i$
1	0	0	-2	$-2A_i$
1	0	1	-1	$-A_i$
1	1	0	-1	$-A_i$
1	1	1	0	0

Step 4 — split the work

The encoding is a function of B alone, so it is done in the dispatcher, which sends the digits d_i^j — the control signals of the selection multiplexers — to the PE; only the selection of the multiple stays inside:

$$\underbrace{\{B_i^{2j+1}, B_i^{2j}, B_i^{2j-1}\} \longrightarrow d_i^j}_{\text{dispatcher: encoding}} \longrightarrow \underbrace{\sum_{j=0}^1 4^j \sum_{i=0}^7 d_i^j A_i}_{\text{PE: selection}}$$

Each d_i^j travels as its 3-bit window $\{B_i^{2j+1}, B_i^{2j}, B_i^{2j-1}\}$, which the cell decodes into one of $\{0, \pm A_i, \pm 2A_i\}$: 6 control bits per lane.

Step 5 — slices of a wider B

A lower nibble L read as signed is short by $16 b_3$; the bit below hands $+b_3$ to the upper nibble H in the DP8 above, and the $\times 16$ of the tree cancels the two, element by element, whatever the grouping of the DP8s:

$$16 (H + b_3) A + (L - 16 b_3) A = (16 H + L) A$$

B is the narrow operand: 2 digits $\times$ 8 lanes = 16 selections of 10 bits per DP8, each over $\{0, \pm A_i, \pm 2A_i\}$.